INTERNATIONAL MONETARY FUND

SIERRA LEONE

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THIRD AND FOURTH REVIEWS UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUEST FOR EXTENSION AND REPHASING OF THE ARRANGEMENT, AND WAIVERS OF NONOBSERVANCE OF PERFORMANCE CRITERIA, AND FINANCING ASSURANCES REVIEW—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By
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Prepared by the staffs of the International Monetary Fund and the International Development Association

Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	No

Sierra Leone's debt is sustainable. This is predicated on the authorities' ambitious fiscal adjustment and continued reliance on concessional financing and grants. The risk of external and overall debt distress remains high. The COVID-19 shock has elevated these risks. The baseline projection reflects the deleterious effects of COVID-19 on growth, exports, and revenues and their gradual recovery after 2021, as well as the cost of measures to counter the health and socioeconomic effects of the pandemic. This DSA also reflects disbursements under the IMF's Rapid Credit Facility (RCF) in 2020 and early 2021, projected future disbursements under the Extended Credit Facility (ECF), debt relief under the IMF's Catastrophe Containment and Relief Trust (CCRT), and debt deferment under the Debt Service Suspension Initiative (DSSI) including the recent extension until end-2021. While the present value of public debt-to-GDP ratio tracks downwards over the medium to long term, it takes about ten years to fall below the indicative thresholds. Some external debt indicators also remain above the thresholds over the medium term. The public debt service-to-revenue ratio and the external debt service-to-exports ratio rise over the medium term, indicating a tight liquidity situation, before steadily declining in the medium to long term. The stress tests highlight the sensitivities to shocks to growth, commodity prices, and exports. Maintaining debt sustainability requires sustained fiscal adjustment underpinned by strengthened public financial management, effective expenditure prioritization, and redoubling structural and revenue mobilization reforms. To adjust at a pace that does not imperil the post-pandemic recovery, and allow for adequate social and priority spending and meeting the country's large development needs, Sierra Leone should continue to rely on highly concessional financing and ideally grants.

[1]The Composite Indicator score of 2.69, based on the April 2021 WEO and the World Bank's latest CPIA, indicates a weak debt-carrying capacity.

PUBLIC DEBT COVERAGE

1. **The DSA covers known sources of public debt** (Text Table 1). As in earlier DSAs, the debt stock includes central government public and publicly-guaranteed debts. The DSA also includes the latest estimate of the consolidated stock of domestic payment arrears. The Government is working, with the support of development partners, to improve its financial and debt management systems, and to enhance the accounting and timely reporting of public debt, including those of state-owned enterprises (SOEs) and self-accounting bodies.

Text Table 1. Public Sector Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

2. **The contingent liability stress test accounts for vulnerabilities associated with SOEs and financial market risks** (Text Table 2). The contingent liability for SOE debt is set at 7 percent of GDP, higher than the default 2 percent of GDP, reflecting the authorities' estimate of indebtedness of SOEs and self-accounting bodies. Contingent liabilities from financial markets are set at the standard minimum value of 5 percent of GDP, which represents the average fiscal cost of a financial crisis in LICs since 1980. The contingent liability of other elements of the general government is set at 0 percent of GDP because the baseline reflects estimated domestic arrears. Overall, total contingent liabilities are estimated at 12 percent of GDP, as in the previous DSA.

Text Table 2. Coverage of the Contingent Liabilities' Stress Test

1. The country's coverage of public debt	The central government plus social security, central bank, government-guaranteed debt		
	Default	**Used for the Analysis**	**Reasons for deviations from the default setting**
2. Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3. SOE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	7.0	Reflect the authorities' estimate of total indebtedness of SOEs.
4. PPP	35 percent of PPP stock	0.0	
5. Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)		12.0	

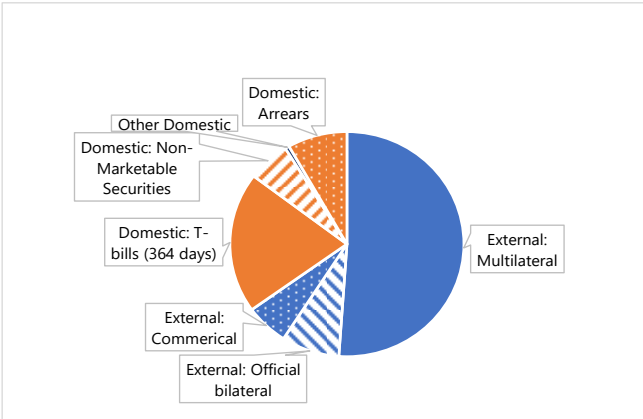
1/ The default shock of 2 percent of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1). If it is already included in the government debt (1) and risks associated with SOE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0 percent.

BACKGROUND ON DEBT

3. The COVID-19 shock seriously strained Sierra Leone's public finances, and total public debt increased in 2020. The large shock to growth and revenues, and measures to counter the impact of the pandemic, increased the fiscal deficit in 2020. Total public debt at end-2020 is estimated to be around 74 percent of GDP, somewhat higher than end-2019. Public debt would fall slightly below this level in 2021 and steadily decrease over the medium term.

4. Public and publicly guaranteed (PPG) external debt increased to around 48 percent of GDP in 2020 from around 44 percent at end-2019. It is expected to increase further in 2021. This reflects additional loans incurred in 2020 and early 2021, namely RCF disbursements in June 2020 (SDR 103.7 million or 50 percent of quota) and in March 2021 (SDR 35.26 million or 17 percent of quota). About 79 percent of Sierra Leone's PPG external debt at end-2020 comprised non-restructurable obligations to multilateral creditors (Text Figure 1). The IMF and World Bank account for about 26 percent and 22 percent of total PPG external debt (US$509 million and US$432 million respectively), followed by the African Development Fund (around 8 percent or US$161 million), the Islamic Development Bank (around 6 percent or US$127 million), the EEC/EIB (around 5 percent or US$90 million). Each of other multilateral creditors accounts for less than 5 percent. Official bilateral creditors account for around 12 percent of total PPG external debt, with the Kuwait Fund the largest (around 3 percent or US$64 million).[2] Sierra Leone has pre-HIPC arrears to external commercial creditors (about 9 percent of PPG external debt or around US$179 million at end-2020).

Text Figure 1. Composition of Public Debt, End-2020



5. Public domestic debt declined to about 26 percent of GDP at end-2020 from around 28 percent of GDP at end-2019. Around three-quarters of domestic debt is owed to commercial banks mainly in the form of 364-day T-bills. Less than a tenth of obligations are to the non-bank sector, while the Bank of Sierra Leone holds less than 15 percent of public debt. The authorities started to clear some domestic arrears balances in the first half of 2020 under their new arrears clearance strategy, while significant external support in 2020 (including the RCF disbursement and budget support grants) helped contain net issuances of relatively more costly T-bills. Roll-over risks are limited, as Sierra Leone has a shallow credit market and domestic banks have limited alternative investment options.

[2]Each of other official bilateral creditors accounts for less than 3 percent of total PPG external debt.

KEY ASSUMPTIONS UNDER THE BASELINE SCENARIO

6. The assumptions are consistent with the macroeconomic framework in the staff report.

- **Growth**. Real GDP in 2020 is estimated to have contracted by 2.2 percent, about a 6½ percentage point decline relative to the pre-COVID projection. This reflects disruptions in production and trade due to the COVID-19 pandemic, as well as the delayed resumption of iron ore mining. Real GDP growth is expected to rebound to about 3.2 percent in 2021, and it is projected to reach its long-run potential of around 4½ percent over the medium term. Downside risks to the growth forecast include a resurgence of the virus including in the rest of the world and a resurging third wave of infections in Sierra Leone.[3] There are both upside and downside risks to mining production.

- **Inflation**. Inflation, as measured by the GDP deflator, is estimated to be broadly in line with average consumer price inflation in 2020 at around 14 percent, and it is projected to decline gradually. Consumer price inflation is projected to reach single digits by 2024, with ongoing efforts to strengthen the monetary policy framework and an easing of fiscal financing pressures.

- **Fiscal**. The overall balance and primary balance are estimated to have widened significantly in 2020 (to -5.6 and -2.6 percent of GDP, respectively) because of the negative shock to domestic revenue and additional crisis related spending. Sustained adjustment in the medium to long term, while protecting critical social and health spending, is consistent with the Government's efforts before the pandemic, in the context of the ECF-supported program, to place the debt-to-GDP ratio on a declining path by the end of the current program. Revenue assumptions reflect the authorities' continued commitment to revenue mobilization, underpinned by revenue administration measures (e.g., ITAS, electronic cash registers to enhance GST compliance, enhanced compliance of selected sectors and large taxpayers) and tax policy reforms, including plans to develop a comprehensive medium-term revenue strategy with support from IMF and World Bank technical assistance. While the fiscal projection provides room for priority social and inclusive spending (thus supportive of growth), scaling up to meet large investment needs would be contingent on progress on revenue mobilization and access to external grants. This DSA reflects additional on-lending from ECF disbursements from the ECF third review onwards—while this does not increase overall exposure to the Fund, it helps reduce reliance on relatively more costly domestic bank financing and improves the composition of financing. Overall financing needs are projected to decline, with continued fiscal adjustment, while domestic borrowing from the banking sector is expected to decline below 2 percent of GDP by 2024.

- **External.** Both exports and imports are estimated to have fallen significantly in 2020, and are projected to recover gradually from 2021 (Table 1). While external budget support grants

[3]As of mid-June 2021, only 1 percent of the population had received their first dose of a COVID-19 vaccine.

increased significantly in 2020, they are anticipated to fall to more normal levels in 2021. The IDA financing is assumed to be all grants until 2031, and 50 percent grants and 50 percent loans after 2032. This DSA reflects two RCF disbursements in June 2020 (SDR 103.7 million or 50 percent of quota) and in March 2021 (SDR 35.26 million or 17 percent of quota), debt relief under the CCRT, and debt deferment under the DSSI.[4] This DSA also takes into account the projected external financing gap during 2023-26, and assumes that the gap will be covered by concessional financing with an overall grant element of 35.5 percent (possibly a combination of grants and loans).[5] This DSA does not reflect the prospective SDR allocation later in 2021.[6]

Text Table 3. Macroeconomic Assumptions

	Current DSA		Previous DSA (March 2021)	
	2021	2031	2021	2030
Real GDP growth (in percent)	3.2	4.5	3.0	4.5
Inflation (GDP deflator, in percent)	11.5	5.1	13.9	5.7
Primary deficit (percent of GDP)	1.1	-2.0	0.7	-2.2
Non-interest current account deficit (percent of GDP)	15.2	11.6	13.1	6.7

7. Arrears clearance and domestic financing. Sierra Leone has a large stock of legacy domestic payment arrears, close to 10 percent of GDP at end-2019. Clearing these arrears is a priority under the authorities' ECF-supported program (Box 1). Developments related to, and the assumptions on, arrears clearance remain unchanged from the previous DSA in March 2021, although they are somewhat less ambitious than the June 2020 DSA in terms of NPV reductions. In the wake of COVID-19, the authorities prioritized speedy clearance to support some small and medium sized enterprises, rather than large NPV reductions envisaged in their arrears clearance strategy.[7] The overall NPV reduction on total stock is therefore on the order of 35-40 percent, as opposed to the 55-60 percent assumed in the June 2020 DSA. The fact that these domestic payment arrears do not accrue interest or charges, and that the authorities' arrears clearance strategy targets significant NPV reductions will be crucial to ensure the cost of clearing arrears is sustainable. The target of gradually reducing domestic bank financing to about 2 percent of GDP remains consistent with the ECF-supported program.

[4]Debt service deferment under the DSSI is projected to be around US$6.6 million in 2020 and around US$17.3 million in 2021 (the amount in 2021 reflects the second extension of the DSSI until the end of 2021). This DSA also reflects the third tranche of the CCRT approved in April 2021 and the corresponding early repayment to the Fund.

[5]More specifically, IMF financing under the ECF has a 5.5-year grace period and 10-year maturity and currently carries a zero interest (the interest rate is subject to Executive Board review every two years). Loans from other multilateral creditors other than the World Bank are assumed to have an overall grant element of 35 percent.

[6]While the SDR allocation does not directly impact PPG external debt, it may influence total public debt depending on associated transactions and arrangements (e.g., net interest payments increase when the holding of SDRs drops below the allocated amount, domestic debt may increase when the central bank on-lends to the government).

[7]The Government paid down about 1.5 percent of GDP in arrears by mid-2020, while the arrears clearance strategy was finalized in June 2020 and approved by the Cabinet in July 2020.

Box 1. Actions to Quantify and Clear Domestic Payment Arrears in Sierra Leone

Resolving the longstanding challenge of domestic expenditure arrears is a key priority for the authorities. In the context of their program, supported by an Extended Credit Facility (ECF) arrangement with the IMF, they have undertaken a comprehensive stocktaking exercise, and adopted an arrears clearance strategy. Given the magnitude of the arrears stock and the tight fiscal financing situation, the clearance strategy reflects the difficult choices, including significant NPV reductions, required to sustainably address this issue. This DSA reflects the current stock of domestic arrears and the planned terms for arrears clearance, which are unchanged from the March 2021 DSA.

The Sierra Leonean authorities have made a concerted effort to understand the magnitude of domestic payment arrears.

- At the time, the ECF-supported program was approved in November 2018, the stock of arrears was estimated at about 4 percent of GDP—reflecting verified (crystalized) but unpaid checks accumulated at the Ministry of Finance—but indications were that it could be higher.
- In early 2019, they published several audits aimed at identifying inherited fiscal risks, including an audit by the Audit Service Sierra Leone (ASSL) to verify claims on the Government, including both future obligations and arrears.[1]
- During June-August 2019, the Ministry of Finance worked with ASSL, and relevant Ministries, Departments and Agencies to review and reconcile individual claims to determine the portion of verified claim in arrears.
- That stocktaking exercise, completed in September 2019, confirmed pre-April 2018 domestic payment arrears of Le 3.3 trillion (around 8¾ percent of 2019 non-iron ore GDP), of which around 90 percent were accrued during 2016 and 2017.[2]
- In addition to these legacy arrears, the authorities have continued to monitor the net accumulation of new arrears (unpaid checks) in the period since April 2018.

The authorities have also finalized their comprehensive arrears clearance strategy. Cognizant of the potentially substantial fiscal implications, the authorities collaborated closely with IMF staff, to develop a strategy to clear arrears based on the following key principles: transparency and equity, sustainability, macrofinancial stability, and preventing further arrears accumulation. The final strategy—approved by Cabinet in July 2020 and published on the *Ministry of Finance* website—implies full clearance over the medium term, assuming deep haircuts or NPV reductions.

The authorities have started clearing arrears at a pace faster than envisaged in the clearance strategy. To cushion the impact of the COVID-19 pandemic on the private sector, and provide liquidity to the banking sector, the authorities prioritized clearing arrears to small and medium enterprises. However, as the haircuts applied were less than envisaged under their clearance strategy, the estimated overall NPV reduction of the prospective arrears clearance is also less than originally envisaged under the plan.

This DSA fully reflects the latest stock of arrears (Le 2.5 trillion as of end-December 2020). Since the September 2019 stocktaking exercise, all Fund-Bank DSAs have included the full stock of legacy arrears (adjusted for arrears that have since been cleared) and the net accummuation of new arrears (unpaid checks) since April 2018, and the expected terms of prospective arrears clearance (as noted above).

[1] See *Box 3. Progress Towards Domestic Arrears Clearance* in IMF Country Report No. 19/217, July 2019.
[2] See *Box 3. Managing and Preventing Expenditure Arrears in Sierra Leone* in IMF Country Report No. 20/116, April 2020.

8. The "realism tool" suggests that the projected fiscal adjustment is realistic (Figure 4). Notwithstanding the impact of the COVID19 shock in delaying the nearer term reduction in domestic bank financing, the longer-term fiscal trajectory remains in line with the pre-crisis level and previous projections under the ECF supported program. The three-year average of fiscal adjustment

over 2021-23 reflects recovery from a sharp deterioration in the primary balance in 2020 due to the COVID19 shock. The projected fiscal adjustment from 2021 onwards is realistic and feasible, considering: (i) expected recovery in domestic revenues as in-person enforcement of tax payments resumes and payoffs from more comprehensive revenue reforms estimated at about 0.1 percent of GDP per year; (ii) the unwinding of emergency spending and gradual efficiency improvements (including in improved appraisal, selection, and execution of priority development projects); and (iii) some fiscal adjustments could be mitigated by the prospective SDR allocation, providing an additional space for critical expenditures of about 0.8 and 0.3 percent of GDP for 2022 and 2023 (Box 1 of ECF 3rd/4th Review staff report).[8]

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

9. Sierra Leone's debt-carrying capacity is assessed to be weak. With the Composite Indicator (CI) score of 2.69 based on the April 2021 WEO and the World Bank's latest CPIA, Sierra Leone's debt carrying capacity is evaluated as weak (Text Table 4). The debt carrying capacity was changed from medium to weak in late 2020, primarily due to weaker global growth projections. Text Table 5 shows applicable thresholds for debt indicators.

Text Table 4. Composite Indicator (CI) Rating

Components	Coefficients (A)	10-year average values (B)	CI Score components	Contribution of
CPIA	0.385	3.161	1.22	45%
Real growth rate (in percent)	2.719	3.734	0.10	4%
Import coverage of reserves (in percent)	4.052	35.427	1.44	53%
Import coverage of reserves^2 (in percent)	-3.990	12.551	-0.50	-19%
Remittances (in percent)	2.022	1.187	0.02	1%
World economic growth (in percent)	13.520	3.078	0.42	15%
CI Score			2.69	100%
CI rating			Medium	

Text Table 5. Applicable Thresholds for Debt Indicators

PV of PPG external debt in percent of GDP	30%
PV of PPG external debt in percent of exports	140%
PPG external debt service in percent of exports	10%
PPG external debt service in percent of revenue	14%
PV of total public debt in percent of GDP	35%

[8] See Sierra Leone—Third and Fourth Review Under the Extended Credit Facility (CR/21/xx)

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

External DSA

10. Sierra Leone is assessed to be at high risk of external debt distress, and PPG external debt is assessed to be sustainable. This assumes strong fiscal adjustment and continued reliance on concessional financing and grants.

- Under the baseline scenario, the PV of PPG external debt-to-GDP ratio and the PV of PPG external debt-to-export ratio exceed their thresholds (30 percent and 140 percent respectively) over the medium term, and these return to their thresholds around 2024-26 (Figure 1, Table 1).[9] The projected decline in the debt ratios reflects several factors such as repayment of past ECF disbursements including those from Ebola period and the projected improvement in GDP and exports. While the historical scenario indicates increasing debt ratios, this reflects the COVID-19 crisis in 2020 and Ebola crisis and commodity price shocks earlier.

- PPG external debt service-to-exports ratio also stays above the threshold of 10 percent over the medium term. In the meantime, PPG external debt service-to-revenue ratio rises over the medium term, indicating a tight liquidity situation, before steadily declining in the medium to long term, while it remains significantly above its threshold of 14 percent for the next ten years.

- Stress tests indicate that the external debt indicators are sensitive to growth and exports. In the stress scenarios, all the external debt indicators remain above the thresholds for the next ten years.

- Since the external debt indicators breach their thresholds under the baseline, Sierra Leone is assessed to be at high risk of external debt distress. However, all the external debt indicators are on a declining trend over the medium- to long-term, and PPG external debt is assessed to be sustainable .

Overall Risk of Public Debt Distress

11. Sierra Leone is assessed to be at high overall risk of public debt distress, and public debt is assessed to be sustainable. Again, this is built on the assumption of strong fiscal adjustment and continued reliance on concessional financing and grants.

- Under the baseline scenario, the PV of public debt-to-GDP ratio gradually declines to the threshold of 35 percent before 2030 (Figure 2). The public debt service-to-revenue ratio is projected to rise over the medium term, suggesting a tight liquidity situation. This increasing

[9]The residuals in Table 1 reflect exchange rate changes.

debt service is expected to be financed with external grants and concessional loans and government revenues. In this context, Sierra Leone will need continued access to concessional financing, including from the Fund, to ensure that financing terms remain contained. In the long term, as the economy fully recovers and revenue mobilization gains materialize, the public debt service-to-revenue ratio is expected to decline gradually.

- Stress tests indicate that the public debt indicators are sensitive to shocks to growth and commodity prices. Considering that both external debt indicators and public debt indicators exceed their thresholds under the baseline, the country is assessed to have high overall risk of public debt distress.

- Nevertheless, public debt is assessed as sustainable given the downward trend in all debt indicators, conditional on (i) sustained and significant fiscal adjustment, and (ii) continued reliance on highly concessional external financing (ideally grants) including from the IFIs which account for a large share of Sierra Leone's PPG external debt, while limiting recourse to expensive domestic debt.

RISK RATING AND VULNERABILITIES

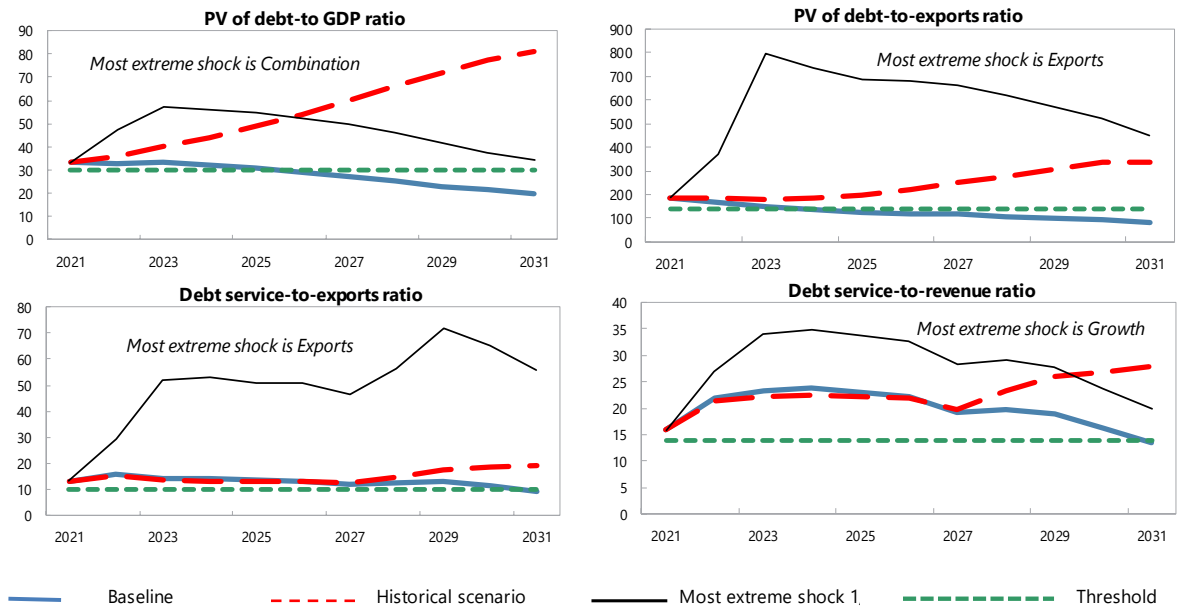
12. While Sierra Leone is assessed to be at high risk of external and overall public debt distress, its debt is assessed to be sustainable. While the COVID-19 shock worsened the public debt situation by weakening growth, revenue, and exports, the medium- to long-term trajectories of debt ratios remain largely unchanged from the pre-pandemic projection. As all the debt indicators remain on a declining trend over the medium to long term, debt is assessed to be sustainable. This is conditional on sustained fiscal adjustment and continued reliance on highly concessional external financing (ideally grants), while limiting recourse to expensive domestic debt. However, the increasing public debt service-to-revenue ratio over the medium term suggests high liquidity-related vulnerabilities. The stress tests also highlight that debt indicators are sensitive to shocks to growth, commodity prices, and exports.

13. This DSA underscores the importance of continued fiscal discipline efforts and structural reforms, supported by technical assistance and prudent financing choices. Reducing debt and maintaining debt sustainability requires, first and foremost, sustained fiscal adjustment, underpinned by strengthened public financial management, effective expenditure prioritization, and redoubling structural and revenue mobilization reform efforts. However, to achieve a pace of fiscal adjustment that does not imperil the post-pandemic recovery and allows the country to protect critical social and health spending and continue addressing its large development needs, it will be vital to rely on highly concessional financing and ideally grants. In line with the limit imposed under the ECF-supported program, Sierra Leone should not contract non-concessional external debt. In addition, continued technical assistance in debt management and development of a domestic market will be important, including an ongoing IMF technical assistance on debt recording and more recently joint World Bank/IMF technical assistance on a medium-term debt strategy.

Authorities' Views

14. The authorities concurred with staff assessment on the risk of debt distress and debt sustainability. The authorities also acknowledged the importance of redoubling efforts to ensure sustained fiscal adjustment and highlighted the range of ongoing reforms to strengthen revenue administration, improve public expenditure management, and further enhance debt management. To safeguard debt sustainability, the authorities committed to continue to prioritize concessional loans within a pre-defined borrowing limit, preferably grants, and seek only highly concessional financing and ratify loans only within the agreed annual ceiling under the ECF-supported program. The authorities also agreed with the need to limit recourse to expensive domestic debt. The authorities reiterated their commitment to clear domestic arrears in line with their arrears clearance strategy. On the external front, the authorities recognize the need to improve the external position and ensure adequate reserve coverage through fiscal adjustment and exchange rate flexibility.

Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2021–31



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	5	5

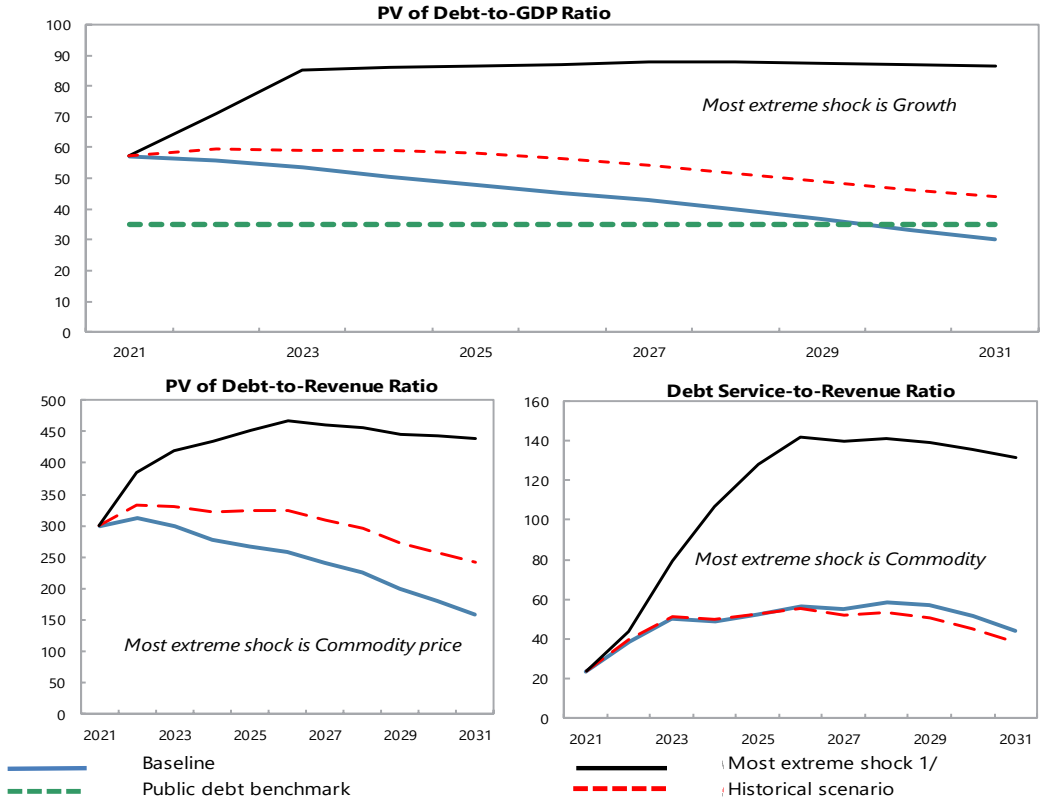
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Sierra Leonean authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (in any), while these are one-breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2021–31



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	33%	33%
Domestic medium and long-term	0%	0%
Domestic short-term	67%	67%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	2%	2%

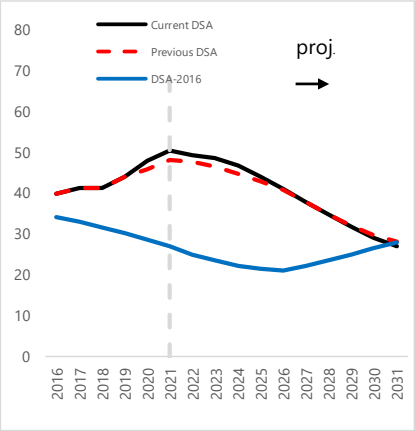
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Sierra Leonean authorities; and IMF staff estimates and projections.

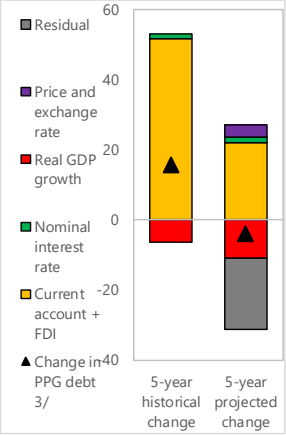
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (in any), while the one-breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Sierra Leone: Drivers of Debt Dynamics, Baseline Scenario, 2016–31

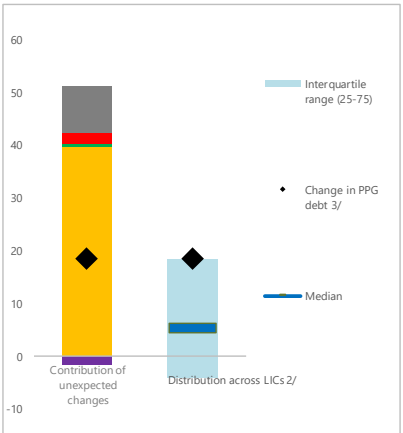
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

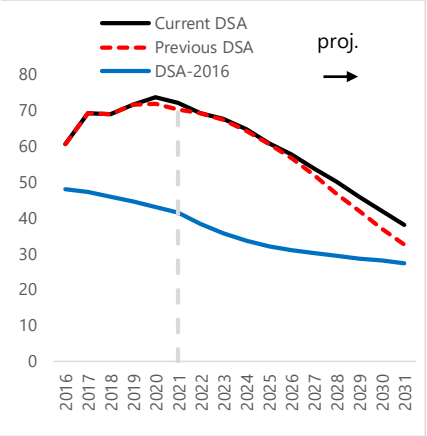


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

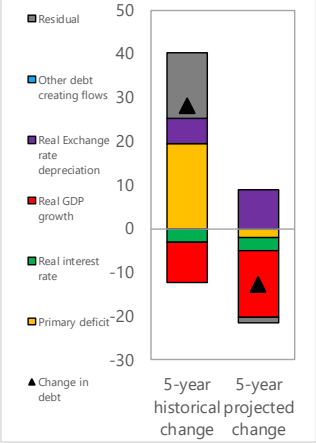


Public debt

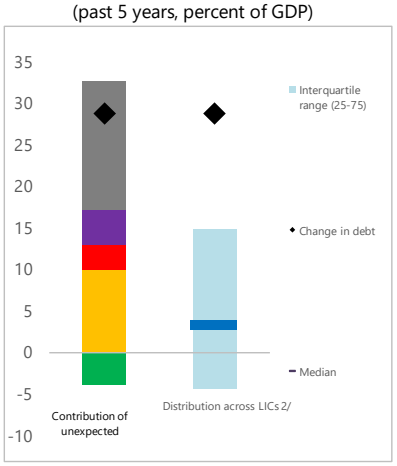
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



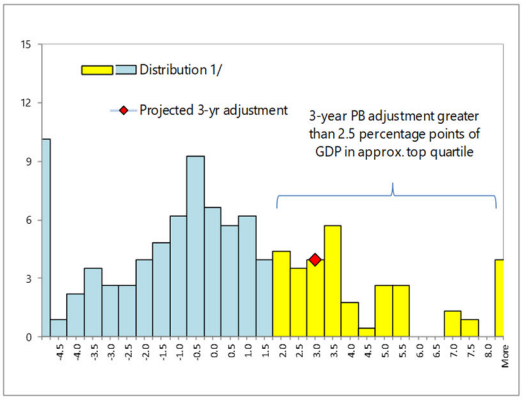
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

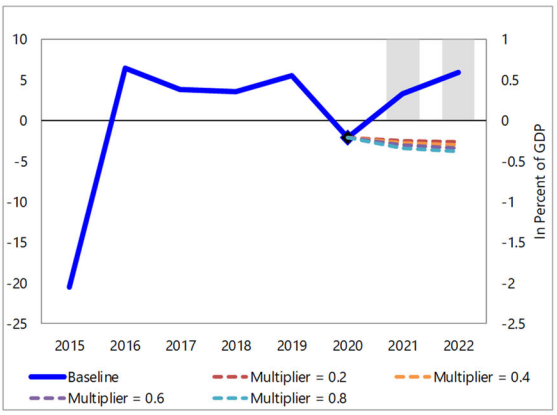
Figure 4. Sierra Leone: Realism Tools

3-Year Adjustment in Primary Balance
(In Percent of GDP)



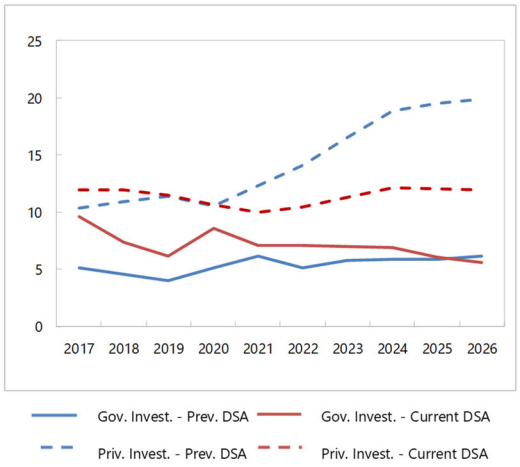
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/
(In Percent)



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

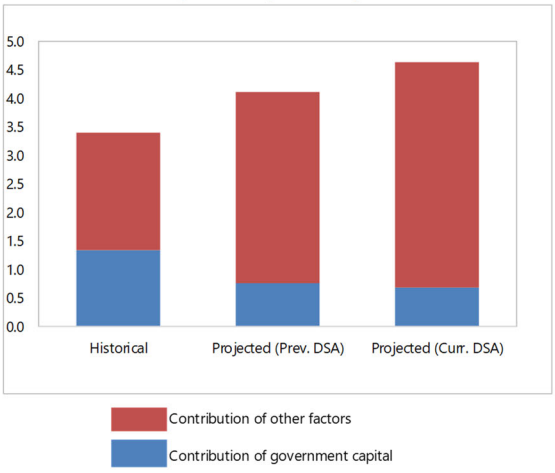
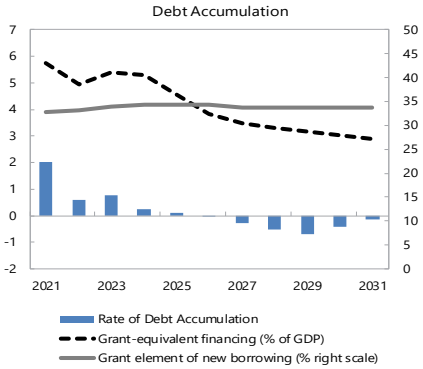


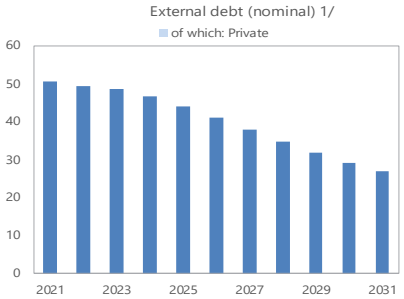
Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2020–41

(Percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**48.0**	**50.5**	**49.3**	**48.7**	**46.7**	**43.9**	**41.1**	**27.1**	**18.5**	**34.9**	**40.1**
of which: public and publicly guaranteed (PPG)	**48.0**	**50.5**	**49.3**	**48.7**	**46.7**	**43.9**	**41.1**	**27.1**	**18.5**	**34.9**	**40.1**
Change in external debt	3.9	2.5	-1.2	-0.6	-2.0	-2.8	-2.8	-2.1	-0.3		
Identified net debt-creating flows	**...**	**4.9**	**3.4**	**2.5**	**1.3**	**0.6**	**0.2**	**-0.6**	**0.8**	**...**	**0.3**
Non-interest current account deficit	**16.4**	**15.2**	**14.9**	**12.6**	**11.5**	**10.8**	**11.1**	**11.6**	**13.7**	**23.1**	**12.3**
Deficit in balance of goods and services	25.5	21.9	20.7	19.0	17.8	16.9	17.0	16.6	17.1	**28.4**	**18.1**
Exports	11.9	17.8	19.6	22.4	24.0	24.8	24.4	24.1	21.5		
Imports	37.4	39.7	40.3	41.3	41.8	41.8	41.4	40.8	38.6		
Net current transfers (negative = inflow)	-10.4	-8.1	-7.2	-7.7	-7.6	-7.5	-7.3	-6.4	-4.7	**-9.3**	**-7.2**
of which: official	-4.7	-3.3	-2.4	-2.9	-2.8	-2.7	-2.6	-2.0	-1.0		
Other current account flows (negative = net inflow)	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	**3.9**	**1.4**
Net FDI (negative = inflow)	**-6.6**	**-9.1**	**-8.8**	**-8.3**	**-8.3**	**-8.4**	**-9.4**	**-11.3**	**-12.3**	**-11.0**	**-10.5**
Endogenous debt dynamics 2/	**...**	**-1.2**	**-2.6**	**-1.8**	**-1.9**	**-1.8**	**-1.4**	**-0.9**	**-0.6**		
Contribution from nominal interest rate	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.2		
Contribution from real GDP growth	0.9	-1.5	-2.9	-2.1	-2.3	-2.1	-1.8	-1.2	-0.8		
Contribution from price and exchange rate changes	...	...	...	...	...	...	...	...	...		
Residual 3/	**...**	**-2.4**	**-4.7**	**-3.1**	**-3.3**	**-3.4**	**-3.1**	**-1.5**	**-1.1**	**...**	**-2.2**
of which: exceptional financing	...	-1.1	-0.3	0.1	0.1	0.1	0.1	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**32.3**	**33.2**	**32.9**	**33.1**	**32.2**	**30.9**	**29.3**	**19.9**	**12.9**		
PV of PPG external debt-to-exports ratio	**271.7**	**186.5**	**167.4**	**147.9**	**134.3**	**124.4**	**120.5**	**82.3**	**59.8**		
PPG debt service-to-exports ratio	**20.3**	**13.1**	**15.6**	**14.2**	**13.9**	**13.3**	**13.3**	**9.3**	**5.6**		
PPG debt service-to-revenue ratio	**17.1**	**15.9**	**22.0**	**23.2**	**23.8**	**23.0**	**22.2**	**13.5**	**6.0**		
Gross external financing need (Million of U.S. dollars)	513.9	366.3	405.9	338.9	306.7	280.9	251.1	179.2	365.7		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-2.2	3.2	5.9	4.4	4.9	4.8	4.2	4.5	4.5	**4.3**	**4.5**
GDP deflator in US dollar terms (change in percent)	4.3	0.3	-3.1	-2.6	-1.4	-0.1	1.0	2.0	2.7	**1.5**	**0.3**
Effective interest rate (percent) 4/	0.8	0.6	0.6	0.7	0.8	0.9	0.9	1.1	1.0	**0.7**	**0.9**
Growth of exports of G&S (US dollar terms, in percent)	-32.6	54.7	13.4	15.8	11.0	8.3	3.3	11.4	6.3	**9.9**	**12.5**
Growth of imports of G&S (US dollar terms, in percent)	-13.7	9.9	4.3	4.2	4.5	4.6	4.4	7.0	6.8	**9.1**	**5.7**
Grant element of new public sector borrowing (in percent)	...	32.9	33.2	33.9	34.3	34.3	34.3	33.7	38.5	**...**	**33.8**
Government revenues (excluding grants, in percent of GDP)	14.2	14.7	13.9	13.7	14.0	14.4	14.6	16.6	20.0	**12.1**	**14.9**
Aid flows (in Million of US dollars) 5/	215.6	191.9	175.7	225.8	240.9	215.7	190.5	162.9	186.2		
Grant-equivalent financing (in percent of GDP) 6/	...	5.7	4.9	5.4	5.3	4.6	3.8	2.9	1.6	**...**	**4.1**
Grant-equivalent financing (in percent of external financing) 6/	...	67.9	70.9	70.1	73.0	70.8	68.4	72.2	58.7	**...**	**71.6**
Nominal GDP (Million of US dollars)	4,201	4,350	4,465	4,539	4,693	4,912	5,173	7,044	13,954		
Nominal dollar GDP growth	2.0	3.5	2.6	1.7	3.4	4.7	5.3	6.7	7.3	**5.9**	**4.8**
Memorandum items:											
PV of external debt 7/	32.3	33.2	32.9	33.1	32.2	30.9	29.3	19.9	12.9		
In percent of exports	271.7	186.5	167.4	147.9	134.3	124.4	120.5	82.3	59.8		
Total external debt service-to-exports ratio	20.3	13.1	15.6	14.2	13.9	13.3	13.3	9.3	5.6		
PV of PPG external debt (in Million of US dollars)	1358.5	1443.0	1468.6	1502.2	1512.9	1518.0	1517.8	1399.0	1796.7		
(PVt-PVt-1)/GDPt-1 (in percent)		2.0	0.6	0.8	0.2	0.1	0.0	-0.1	0.6		
Non-interest current account deficit that stabilizes debt ratio	12.5	12.7	16.1	13.2	13.5	13.6	13.9	13.7	14.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

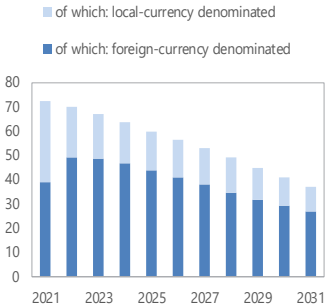
Table 2. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020–41

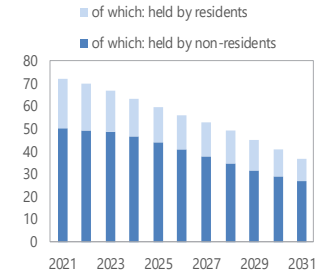
(Percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**73.7**	**72.1**	**70.1**	**67.1**	**63.5**	**59.9**	**56.3**	**37.0**	**20.6**	**53.4**	**55.8**
of which: external debt	**48.0**	**50.5**	**49.3**	**48.7**	**46.7**	**43.9**	**41.1**	**27.1**	**18.5**	**34.9**	**40.1**
Change in public sector debt	2.1	-1.6	-2.0	-3.0	-3.5	-3.6	-3.6	-3.9	-0.5		
Identified debt-creating flows	**-0.1**	**0.5**	**-2.6**	**-1.9**	**-3.4**	**-4.1**	**-3.9**	**-3.8**	**-4.7**	**1.2**	**-3.1**
Primary deficit	**2.6**	**1.1**	**-0.1**	**-0.5**	**-1.2**	**-1.4**	**-1.4**	**-2.0**	**-3.8**	**3.2**	**-1.1**
Revenue and grants	19.3	19.1	17.9	17.9	18.3	18.0	17.5	18.9	20.9	**15.8**	**18.2**
of which: grants	5.1	4.4	3.9	4.2	4.3	3.6	2.9	2.3	0.9		
Primary (noninterest) expenditure	21.9	20.2	17.8	17.4	17.1	16.6	16.1	16.9	17.1	**19.0**	**17.1**
Automatic debt dynamics	**-2.8**	**-0.6**	**-2.5**	**-1.5**	**-2.2**	**-2.7**	**-2.6**	**-1.8**	**-1.0**		
Contribution from interest rate/growth differential	0.5	-3.9	-4.5	-3.4	-3.5	-3.1	-2.5	-1.8	-1.0		
of which: contribution from average real interest rate	-1.1	-1.6	-0.4	-0.4	-0.3	-0.2	-0.1	0.0	-0.1		
of which: contribution from real GDP growth	1.6	-2.3	-4.0	-2.9	-3.1	-2.9	-2.4	-1.8	-0.9		
Contribution from real exchange rate depreciation	-3.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.2**	**1.2**	**2.5**	**0.8**	**1.1**	**0.9**	**0.3**	**-0.1**	**4.3**	**1.5**	**0.6**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**59.0**	**57.3**	**56.0**	**53.4**	**50.7**	**48.0**	**45.3**	**30.1**	**15.3**		
PV of public debt-to-revenue and grants ratio	**306.0**	**300.2**	**313.1**	**298.7**	**276.6**	**267.0**	**258.6**	**159.3**	**73.3**		
Debt service-to-revenue and grants ratio 3/	**25.4**	**23.6**	**38.7**	**50.2**	**48.6**	**51.9**	**56.4**	**43.9**	**6.5**		
Gross financing need 4/	7.5	5.6	6.8	8.5	7.7	8.0	8.5	6.3	-2.4		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-2.2	3.2	5.9	4.4	4.9	4.8	4.2	4.5	4.5	**4.3**	**4.5**
Average nominal interest rate on external debt (in percent)	0.8	0.7	0.7	0.8	0.8	0.9	0.9	1.1	1.0	**0.8**	**0.9**
Average real interest rate on domestic debt (in percent)	-3.3	-1.7	1.0	1.6	1.8	2.2	2.3	1.8	2.1	**0.1**	**1.6**
Real exchange rate depreciation (in percent, + indicates depreciation)	-7.2	...	...	...	...	...	...	...	...	**0.7**	**...**
Inflation rate (GDP deflator, in percent)	13.8	11.5	11.8	10.5	9.5	8.3	6.9	5.1	4.8	**10.9**	**7.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	16.7	-5.1	-6.6	2.3	3.0	1.6	1.3	5.9	4.5	**6.2**	**2.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.6	2.7	1.9	2.6	2.3	2.3	2.2	2.0	-3.3	**0.4**	**2.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Note: Grants include grants for debt service relief under the Catastrophe Containment and Relief Trust (CCRT), projected to amount to 0.7 percent of GDP in 2021.

Table 3. Sierra Leone: Sensitivity Analysis, External Debt, 2021-2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	**33.2**	**32.9**	**33.1**	**32.2**	**30.9**	29.3	27.5	25.4	23.2	21.3	19.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**33.2**	**35.9**	**39.9**	**44.2**	**48.8**	**53.8**	**60.0**	**66.1**	**71.7**	**77.2**	**81.2**
B. Bound Tests											
B1. Real GDP growth	**33.2**	**40.4**	**48.6**	**47.3**	**45.4**	**43.1**	**40.3**	**37.2**	**34.0**	**31.3**	29.1
B2. Primary balance	**33.2**	**34.1**	**36.5**	**37.0**	**36.5**	**35.5**	**33.9**	**31.8**	29.4	27.3	25.5
B3. Exports	**33.2**	**41.2**	**54.7**	**54.0**	**52.5**	**50.6**	**48.4**	**45.2**	**40.7**	**36.8**	**33.5**
B4. Other flows 3/	**33.2**	**44.0**	**55.6**	**55.0**	**53.7**	**51.9**	**49.8**	**46.3**	**41.6**	**37.5**	**33.9**
B5. One-time 30 percent nominal depreciation	**33.2**	**42.0**	**37.7**	**36.5**	**34.9**	**32.9**	**30.6**	27.9	25.6	23.8	22.3
B6. Combination of B1-B5	**33.2**	**47.6**	**57.1**	**56.2**	**54.5**	**52.5**	**50.0**	**46.1**	**41.6**	**37.7**	**34.3**
C. Tailored Tests											
C1. Combined contingent liabilities	**33.2**	**35.6**	**37.6**	**37.8**	**37.2**	**36.1**	**34.4**	**32.4**	**30.2**	28.3	26.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**33.2**	**34.2**	**35.8**	**34.9**	**33.3**	**31.3**	28.8	25.9	22.8	20.2	18.0
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	**186.5**	**167.4**	**147.9**	134.3	124.4	120.5	115.2	106.4	99.1	92.2	82.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**186.5**	**182.7**	**178.5**	**183.9**	**196.4**	**221.1**	**251.7**	**277.1**	**306.7**	**333.8**	**336.4**
B. Bound Tests											
B1. Real GDP growth	**186.5**	**167.4**	**147.9**	134.3	124.4	120.5	115.2	106.4	99.1	92.2	82.3
B2. Primary balance	**186.5**	**173.7**	**162.9**	**154.1**	**147.1**	**145.7**	**142.2**	133.6	126.0	118.2	105.8
B3. Exports	**186.5**	**370.5**	**796.8**	**732.4**	**688.6**	**677.5**	**661.3**	**617.1**	**567.9**	**519.2**	**451.6**
B4. Other flows 3/	**186.5**	**224.0**	**248.7**	**229.2**	**216.1**	**213.2**	**208.9**	**194.1**	**178.2**	**162.3**	**140.5**
B5. One-time 30 percent nominal depreciation	**186.5**	**167.4**	131.9	119.2	109.9	105.8	100.4	91.8	85.9	80.5	72.4
B6. Combination of B1-B5	**186.5**	**286.0**	**207.6**	**343.6**	**322.3**	**316.4**	**307.9**	**283.6**	**261.3**	**239.4**	**208.9**
C. Tailored Tests											
C1. Combined contingent liabilities	**186.5**	**181.0**	**167.9**	**157.6**	**149.8**	**148.1**	**144.3**	135.8	129.1	122.3	110.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**186.5**	**185.0**	**167.2**	**150.3**	137.5	130.7	122.0	109.7	98.5	88.1	75.4
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	**13.1**	**15.6**	**14.2**	**13.9**	**13.3**	**13.3**	**12.1**	**12.5**	**12.8**	**11.2**	9.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**13.1**	**15.1**	**13.5**	**13.2**	**12.8**	**13.1**	**12.4**	**14.7**	**17.5**	**18.7**	**19.1**
B. Bound Tests											
B1. Real GDP growth	**13.1**	**15.6**	**14.2**	**13.9**	**13.3**	**13.3**	**12.1**	**12.5**	**12.8**	**11.2**	9.3
B2. Primary balance	**13.1**	**15.6**	**14.3**	**14.1**	**13.6**	**13.6**	**12.4**	**13.3**	**14.5**	**13.3**	**11.5**
B3. Exports	**13.1**	**29.2**	**52.0**	**53.2**	**51.0**	**50.8**	**46.3**	**56.3**	**72.0**	**65.0**	**55.6**
B4. Other flows 3/	**13.1**	**15.6**	**15.0**	**15.3**	**14.6**	**14.6**	**13.3**	**18.1**	**22.6**	**20.5**	**17.6**
B5. One-time 30 percent nominal depreciation	**13.1**	**15.6**	**14.2**	**13.7**	**13.1**	**13.1**	**11.9**	**12.3**	**11.2**	9.7	7.9
B6. Combination of B1-B5	**13.1**	**20.8**	**26.8**	**26.5**	**25.4**	**25.4**	**23.1**	**30.5**	**33.5**	**30.1**	**25.6**
C. Tailored Tests											
C1. Combined contingent liabilities	**13.1**	**15.6**	**14.4**	**14.2**	**13.7**	**13.7**	**12.4**	**12.8**	**13.2**	**11.6**	9.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**13.1**	**16.8**	**15.2**	**15.0**	**14.3**	**14.2**	**12.8**	**13.6**	**14.3**	**12.4**	**10.1**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	**15.9**	**22.0**	**23.2**	**23.8**	**23.0**	**22.2**	**19.3**	**19.8**	**19.0**	**16.1**	13.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**15.9**	**21.3**	**22.1**	**22.5**	**22.1**	**21.9**	**19.8**	**23.4**	**26.0**	**26.8**	**27.8**
B. Bound Tests											
B1. Real GDP growth	**15.9**	**27.0**	**34.1**	**34.9**	**33.7**	**32.6**	**28.3**	**29.0**	**27.9**	**23.6**	**19.8**
B2. Primary balance	**15.9**	**22.0**	**23.4**	**24.1**	**23.5**	**22.8**	**19.8**	**21.1**	**21.5**	**19.1**	**16.8**
B3. Exports	**15.9**	**23.2**	**26.1**	**27.9**	**27.0**	**26.0**	**22.7**	**27.4**	**32.8**	**28.6**	**24.8**
B4. Other flows 3/	**15.9**	**22.0**	**24.5**	**26.2**	**25.3**	**24.4**	**21.3**	**28.8**	**33.6**	**29.4**	**25.7**
B5. One-time 30 percent nominal depreciation	**15.9**	**28.1**	**29.7**	**29.8**	**28.9**	**27.9**	**24.2**	**24.9**	**21.3**	**17.7**	**14.7**
B6. Combination of B1-B5	**15.9**	**24.8**	**29.8**	**30.9**	**29.9**	**28.8**	**25.1**	**33.1**	**33.8**	**29.3**	**25.4**
C. Tailored Tests											
C1. Combined contingent liabilities	**15.9**	**22.0**	**23.5**	**24.2**	**23.6**	**22.8**	**19.9**	**20.4**	**19.6**	**16.6**	**14.0**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**15.9**	**25.7**	**27.6**	**28.5**	**26.7**	**25.0**	**21.0**	**21.4**	**21.1**	**17.6**	**14.5**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Sierra Leone: Sensitivity Analysis, Public Debt, 2021-2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
	PV of Debt-to-GDP Ratio										
Baseline	**57.3**	**56.0**	**53.4**	**50.7**	**48.0**	**45.3**	**42.9**	**40.1**	**36.7**	33.4	30.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**57**	**60**	**59**	**59**	**58**	**56**	**54**	**51**	**49**	**46**	**44**
B. Bound Tests											
B1. Real GDP growth	**57**	**71**	**85**	**86**	**87**	**87**	**88**	**88**	**87**	**87**	**86**
B2. Primary balance	**57**	**61**	**63**	**60**	**56**	**53**	**51**	**47**	**44**	**40**	**37**
B3. Exports	**57**	**62**	**71**	**69**	**66**	**63**	**60**	**56**	**51**	**46**	**41**
B4. Other flows 3/	**57**	**68**	**77**	**75**	**72**	**68**	**66**	**61**	**56**	**50**	**44**
B5. One-time 30 percent nominal depreciation	**57**	**59**	**56**	**52**	**49**	**45**	**42**	**38**	34	31	27
B6. Combination of B1-B5	**57**	**62**	**63**	**56**	**54**	**51**	**49**	**46**	**43**	**40**	**36**
C. Tailored Tests											
C1. Combined contingent liabilities	**57**	**67**	**63**	**60**	**57**	**53**	**51**	**48**	**44**	**41**	**37**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**57**	**62**	**67**	**72**	**76**	**78**	**80**	**81**	**82**	**83**	**83**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
	PV of Debt-to-Revenue Ratio										
Baseline	300.2	313.1	298.7	276.6	267.0	258.6	241.2	226.0	200.5	179.8	159.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	300	332	330	321	324	324	309	296	273	257	241
B. Bound Tests											
B1. Real GDP growth	300	379	432	426	443	463	462	465	450	442	433
B2. Primary balance	300	342	354	326	314	304	284	268	239	217	194
B3. Exports	300	350	399	375	366	358	337	317	279	248	217
B4. Other flows 3/	300	380	432	407	398	391	369	346	303	268	235
B5. One-time 30 percent nominal depreciation	300	337	318	291	276	261	239	220	190	166	142
B6. Combination of B1-B5	300	346	344	299	295	290	273	259	232	211	190
C. Tailored Tests											
C1. Combined contingent liabilities	300	374	353	326	315	305	286	269	241	219	197
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	300	385	419	435	453	468	460	457	446	443	438
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Debt Service-to-Revenue Ratio										
Baseline	23.6	38.7	50.2	48.6	51.9	56.4	55.2	58.1	56.9	51.4	43.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	24	39	51	50	52	55	52	53	51	45	38
B. Bound Tests											
B1. Real GDP growth	24	45	77	90	107	124	128	138	139	137	131
B2. Primary balance	24	39	71	83	75	72	66	66	64	59	51
B3. Exports	24	39	51	50	53	58	56	63	66	60	52
B4. Other flows 3/	24	39	51	50	54	58	57	66	69	63	55
B5. One-time 30 percent nominal depreciation	24	39	51	49	53	57	55	58	56	51	43
B6. Combination of B1-B5	24	40	53	52	60	68	68	73	72	67	60
C. Tailored Tests											
C1. Combined contingent liabilities	24	39	94	77	71	70	64	64	61	55	47
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	24	44	80	107	128	142	140	141	139	136	131
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.